FERNANDO QUIJADA

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EDUCATION

Duke University – Durham, NC **Expected Graduation: May 2027**

B.S. in Biomedical Engineering; Certificate in Innovation & Entrepreneurship

- GPA: 3.4/4.0; Honors/Programs: A. James Clark Scholar, Duke BME Design Fellow
- Relevant Courses: Intro to Medical Instrumentation, MedTech Prototyping, Signals & Systems, Engineering Design & Communications, Biomedical Electronic Measurement, Modern Diagnostic Imaging

RELEVANT EXPERIENCE

Duke BME Design Fellows Program – Design Fellow, Duke University **May 2026 – July 2026**

- Partnered with a team of 4 and two clients (a breast cancer researcher and an ophthalmologist) to scope two device projects, turning client interviews into prioritized user needs and quantified specifications with minimum, nominal, and ideal targets
- Designed and built a mounting mechanism for a portable LED microscope, iterating through low-fidelity prototypes and repeated client meetings that surfaced missed requirements (mouse access, viewing angle) and drove a complete redesign
- Built a fully mechanical lock-translate-lock mechanism that cut configuration-change time from 2–3 minutes to under 5 seconds, with 3D translation and rotation, mouse clearance, and full compatibility with the client's setup

Outreach Design Education (ODE) Summer Design Camp – Teaching Assistant, Duke University **June 2026 – July 2026**

- Managed 14 Ultimaker S3 printers for a 5-week, 28-student design program; troubleshot nozzle clogs and filament feed failures on 8 printers, tracing recurring jams to fused filament sections to keep students printing ahead of final presentations
- Coached 28 students through the design process and Onshape CAD, taking concepts from sketch to 3D-printed prototype; ran all early slicing and printing, then trained students to slice independently

Deep Learning for Materials Science – Data+ Intern, Data+/Climate+ **May 2024 – July 2024**

- Co-built a Python pipeline that used experimental statistics to synthesize training data across two materials with different crystal structures, standardizing image format and a traceable naming convention to prepare the data for model training
- Trained and benchmarked 8 PyTorch DeepLabv3 models across dataset sizes from 500 to 10,000 images to remove background artifacts from scattering data, identifying that performance gains plateaued beyond 5,000 images due to model limitations rather than data volume

PROJECTS

Pacemaker Design & Characterization – Duke University **Spring 2026**

- Designed and built the analog circuitry for a functional pacemaker prototype, including an instrumentation amplifier and bandpass filter for ECG acquisition, a comparator stage for R-wave detection, and a MOSFET-based stimulation circuit delivering an output current at twice the measured rheobase for reliable capture
- Collaborated with a partner to verify the complete system on a live frog heart model, confirming reliable R-wave detection, under 1% error in calculated heart rate across physiological rates, and correct rate-responsive pacing when a temperature-based rhythm threshold was exceeded

Hearing Aid Design – Duke University **Sprint 2025**

- Designed and soldered a compact hearing aid circuit with microphone input and headphone output, using cascaded high-pass and low-pass op-amp filter stages (LF351) and an LM386 output amplifier with potentiometer-based gain control
- Root-caused a non-functional circuit by isolating and testing each component individually, replacing a faulty potentiometer to restore functionality, then validated the frequency response via Bode plot analysis, confirming it met specs (≥3dB; 75Hz–10kHz)

Lightbox (LED Control System) – Duke University **Spring 2025**

- Designed a themed, 3D-printed enclosure (Onshape) with internal routing for wiring and snap-fit assembly of moving parts, and a custom PCB (KiCad) that was fabricated and hand-soldered to fit inside it
- Built Arduino firmware implementing a state machine that controls LED brightness (0–100% PWM duty cycle)/blink rate (5–10Hz), then statistically validated performance (SciPy, Pandas), confirming accuracy within 0.4Hz and 1.3% with a near-ideal linear response (R^2>0.998)

LEADERSHIP & INVOLVEMENT

Society of Hispanic Professional Engineers – Vice President, Duke University **May 2024 – Present**

- Built the chapter's corporate sponsorship package and led outreach efforts, securing events with Lenovo, Deloitte, and Google
- Founded mentorSHPE, an underclassman–upperclassman program, scaling participation from 4 to 17 pairs chapter-wide
- Directed professional/community programming (resume/interview workshops, NSBE/SWE collaborations) for ~70 members

Durham AIM – High School Mentor, Duke University **August 2025 – December 2025**

- Mentored 8 high school juniors through weekly 30-minute sessions on time management and academic planning, helping all 8 finalize course plans, and compiled resource guides on college summer programs and scholarship opportunities

SKILLS

- CAD & Hardware: Onshape, Fusion 360, 3D printing, PCB design/fabrication (KiCad), oscilloscope, soldering, embedded systems, Arduino
- Programming & Data: Python, PyTorch, NumPy, C/C++, Git, statistical analysis, Bode/transfer function analysis